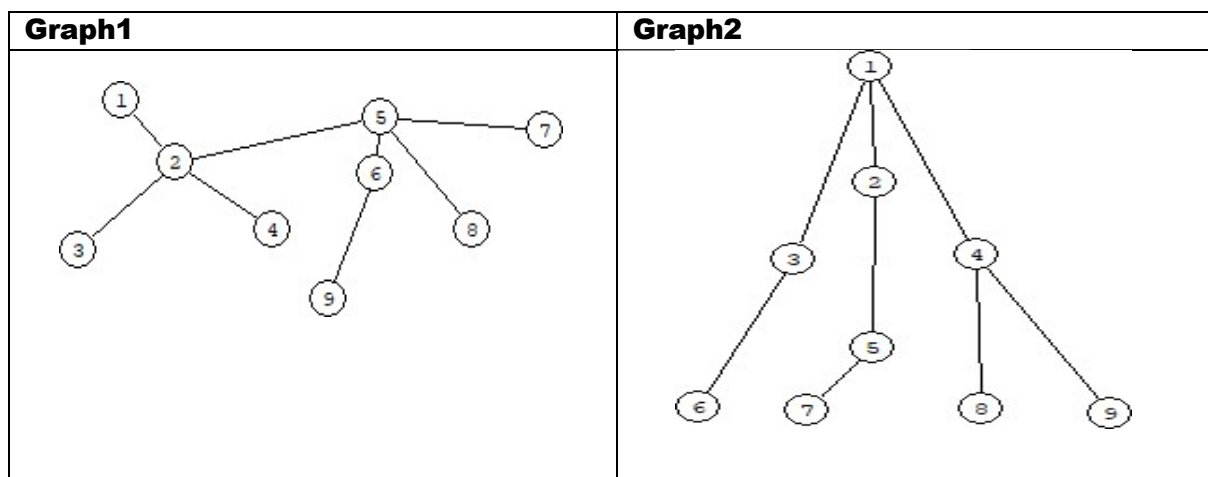


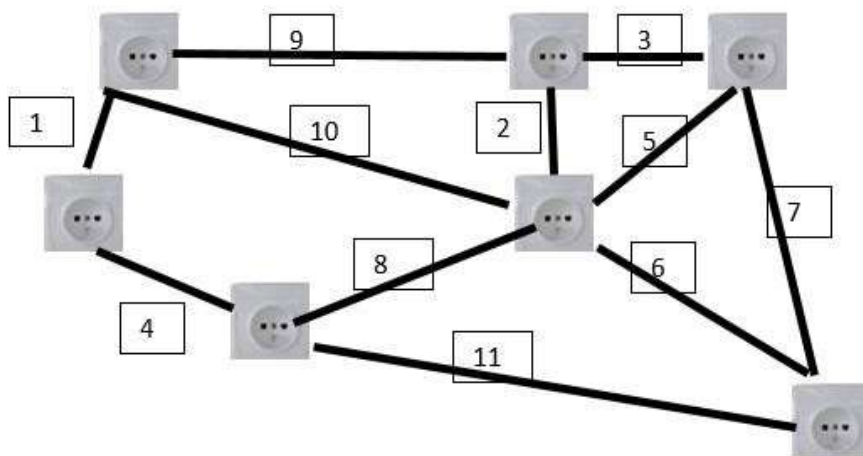
Exercise1:

1. Indicate for each graph the number of edges and the number of isthmuses.
2. What is the type of these graphs?
3. What do you notice?
4. Modify these graphs to make them *rooted trees*.



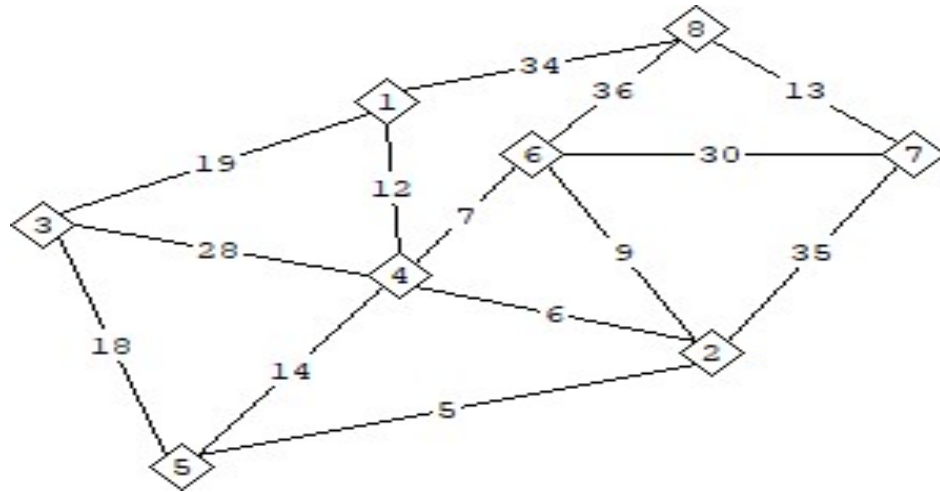
Exercise2:

1. Choose the shortest set of cables to connect all the sockets (the electrical wire is expensive per meter and we want to minimize the installation price).
2. Calculate the length of the electrical wire used.



Exercise3:

1. Extract the minimum spanning tree according to Kruskal then according to Prim.
2. Deduce the weight of the tree.



Exercise 4:

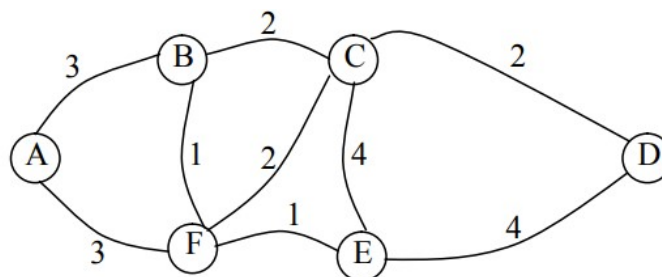
Consider a set of **5 cities**. The cost of building a road between city **i** and city **j** is a_{ij} ; the values of a_{ij} are given in the following matrix $A = [a_{ij}]$:

$$A = \begin{bmatrix} 0 & 3 & 5 & 11 & 9 \\ 3 & 0 & 3 & 9 & 8 \\ 5 & 3 & 0 & +\infty & 10 \\ 11 & 9 & +\infty & 0 & 7 \\ 9 & 8 & 10 & 7 & 0 \end{bmatrix}$$

Find the minimum cost road network that will connect all these cities.

Exercise 5:

A project to install water pipes supplying the localities **A, B, C, D, E** and **F** is represented by the following graph $G=(X,U)$. The values on the edges represent the length of the pipes used for the installation.



Provide an installation plan that minimizes the total length of water pipes used?